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Do Psychiatric Disorders From Late Childhood Through Adolescence Predict School Outcomes Over and Above Temperament? Findings From a Longitudinal Study of Mexican-Origin Youth

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School success is pivotal for minoritized youth from socioeconomically disadvantaged backgrounds to achieve upward social mobility. Using data from the California Families Project, a longitudinal study of 674 Mexican-origin youth living in the United States, the present study examined the association between trajectories of psychiatric disorders from late childhood to adolescence and a wide range of high school outcomes assessed via school records. Psychiatric disorders were examined at different levels of breadth, including individual psychiatric disorders, internalizing symptoms, externalizing symptoms, and total symptoms. Results of latent growth curve models showed that youth exhibiting more overall psychiatric symptoms had lower high school grade point averages and achievement test scores, took fewer challenging courses during high school, and were less likely to attend college compared with youth exhibiting fewer symptoms. Counterintuitively, faster increases (or slower decreases) in psychiatric symptoms were associated with more positive school outcomes. These findings were similar for boys and girls and for youth born in the United States and Mexico. Incremental validity analyses showed that temperament levels were more robust predictors of school outcomes than psychiatric disorder levels. Developmentally, changes in temperament and psychiatric disorders independently predicted school outcomes. Overall, temperament levels and slopes had incremental validity over psychiatric disorder levels and slopes, suggesting that temperament may be a better target of intervention efforts to promote school success than psychiatric symptoms.

Public Significance Statement
Using data from a longitudinal study of 674 Mexican-origin youth, we found that both psychiatric disorders and temperament trajectories from late childhood through adolescence predict a wide range of high school outcomes. Temperament was a more robust predictor of school outcomes, suggesting that it may be a better target of intervention efforts to promote school success than psychiatric disorders.

Keywords: psychiatric disorders, school outcomes, academic achievement, temperament, incremental validity

Supplemental materials: https://doi.org/10.1037/dev0002175.supp

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High school completion represents an educational milestone that fundamentally shapes life trajectories, particularly for socioeconomically disadvantaged minoritized youth such as Mexican-origin students. High school success serves as the essential gateway to higher education—the primary pathway for developing advanced cognitive skills (Van Damme & Zahner, 2022), accessing high-wage career trajectories (Starr et al., 2024), and extending social networks (Vallejo & Ramirez, 2023) to achieve upward social mobility. Beyond its function as a gateway to college, high school academic achievement predicts more favorable life outcomes,

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Study materials and R scripts for this project are publicly available on the Open Science Framework at https://osf.io/8ydqt/?view_only=8f1195a6e8cd45f88f498dd1be7945d8. The authors are not legally or ethically allowed to publicly post data for this project because the participants in the study have not given informed consent to have their personal data publicly shared, and the authors do not have institutional review board approval to post data publicly. Findings of the present article were presented as a poster at the Association for Research in Personality Biennial Conference (2023) and the Society for Personality and Social Psychology's Annual Convention (2024). This research was supported by the National Institute on Drug Abuse (Grant DA017902) and the National Institute of Mental Health (Grant MH123530), awarded to Richard W. Robins. The authors thank the California Department of Education for providing data on school outcomes.

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including higher occupational prestige, higher well-being, and a lower likelihood of poverty in early adulthood (Starr et al., 2024). Given the significant impact of high school academic achievement in breaking the cycle of intergenerational poverty, it is crucial to understand the factors that facilitate or hinder high school achievement among these youth. The present study uses data from a longitudinal study of Mexican-origin youth living in the United States to examine how trajectories of psychiatric disorders from late childhood to adolescence predict success in high school.

High school success is influenced by numerous factors at multiple layers of the ecological system, ranging in their proximity to the individual. At the individual level, psychiatric disorders have received particular attention, as early intervention could minimize their adverse impact on school outcomes. At the societal level, Mexican-origin youth are disproportionally affected by systematic and structural barriers, including low socioeconomic status (Irwin et al., 2022), neighborhood distress (Crowder & South, 2003), underresourced schools (Baker et al., 2024), acculturation stress, and ethnic discrimination. These contextual risks hinder academic achievement and educational attainment both directly (e.g., Civitillo et al., 2024; Liu et al., 2022; Nieuwenhuis & Hooimeijer, 2016) and indirectly by increasing risk for psychiatric disorders. Unfortunately, Hispanic communities face significant disparities in mental health care access compared with non-Hispanic White populations (Rodgers et al., 2022), a gap that widened during the COVID-19 pandemic (Thomeer et al., 2023). These findings highlight the urgent need to address these disparities and develop comprehensive, culturally compatible intervention programs that meet the unique needs of Hispanic communities.

The present study has two aims: (a) examine whether psychiatric disorders from late childhood through adolescence predict educational outcomes (e.g., grades and test scores) in late adolescence and early adulthood and (b) examine whether psychiatric disorders predict educational outcomes over and above temperament traits (effortful control, negative emotionality, surgency, and affiliation) given their substantial conceptual overlap and similar predictive validity for school outcomes. Below, we first discuss prior longitudinal research on the association between psychiatric disorders and academic outcomes and then discuss teasing apart the effects of psychopathology and temperament on educational outcomes.

Psychiatric Disorders and School Outcomes

In the following sections, we review theory and research linking a wide range of psychiatric disorders to educational outcomes. In each section, we first discuss the mechanisms through which each disorder might lead to problems in school and then discuss longitudinal research testing the theorized associations. Previous studies were conducted with predominantly European-background samples. We note longitudinal studies focusing on Latinx youth. Given that psychiatric disorders can be classified hierarchically into a set of continuous dimensions, as specified by the hierarchical taxonomy of psychopathology model (Kotov et al., 2021), we organize our review of prior research into disorders related to the internalizing spectrum (major depressive disorder [MDD], generalized anxiety disorder [GAD], fear-related syndromes and disorders [social phobia, panic disorder, and agoraphobia], and posttraumatic stress disorder [PTSD]) and disorders related to the externalizing spectrum (attention-deficit/hyperactivity disorder [ADHD], oppositional defiant disorder [ODD], conduct disorder [CD], and substance use [including alcohol, marijuana, nicotine, and other substances]). In the present study, we will report findings at three different levels of breadth: individual psychiatric disorders, the internalizing and externalizing spectra, and all disorders (i.e., the general factor of psychopathology, or *p* factor; Caspi et al., 2024). Many of the ideas presented below draw on frameworks, such as the adjustment erosion hypothesis, postulating that symptoms of psychiatric disorders lead to academic underachievement by disrupting learning processes, reducing classroom participation, creating problems in teacher and peer relationships, and increasing the likelihood of deviant peer affiliation (Moilanen et al., 2010).

Associations Between Internalizing Disorders and School Outcomes

The Internalizing Spectrum

The broad category of internalizing disorders, which includes symptoms of depressive and anxiety disorders, is prospectively associated with academic underachievement (for a meta-analysis, see Riglin et al., 2014). Below, we summarize prior findings about how each specific internalizing disorder predicts school outcomes.

MDD

MDD is associated with deficits in cognitive, motivational, and interpersonal functioning that impede learning and performance in school. Cognitively, meta-analyses show that MDD is associated with deficits in executive functions, attention, memory, and intelligence quotient scores, which could lead to poor test scores and grades (Rock et al., 2014; Wagner et al., 2015). Motivationally, youth with higher levels of depression (compared with those with lower levels) tend to be less responsive to rewards and incentives and have weaker approach motivation (Shankman et. al., 2007; Treadway et al., 2012) and therefore are less likely to exert effort on assignments, contribute to group projects, seek help from peers and teachers, and study for upcoming exams. Interpersonally, depressed individuals tend to experience more negative affect in response to social rejection (Silk et al., 2014), impaired social emotion recognition (Dalili et al., 2015), diminished social cooperativeness (Clark et al., 2013), and reduced empathy (Kupferberg et al., 2016). These factors might hinder student–peer and student–teacher relationships. Students who receive less support from peers and teachers (compared with those with more support) tend to have a lower sense of belonging in the school context (Allen et al., 2018), which is detrimental to academic motivation (Neel & Fuligni, 2013), school performance (Korpershoek et al., 2019), and mental health (Allen et al., 2024), particularly among students from marginalized racial–ethnic groups (Walton & Cohen, 2011). Overall, depression symptoms engender cognitive, motivational, and interpersonal processes that contribute to poor school outcomes.

Consistent with this conceptual framework, prior research has shown that depression is linked to poor educational outcomes. A meta-analysis of longitudinal studies with two-wave designs found that depression was prospectively associated with a higher likelihood of school dropout and worse school grades during childhood and adolescence (Riglin et al., 2014). More recent meta-analyses found that depression was prospectively associated with school

dropout (Clayborne et al., 2019) and negatively associated with subsequent grades and test scores (Wickersham et al., 2021).

GAD and Other Anxiety Disorders

GAD and other anxiety disorders (PTSD, social phobia, panic disorder, and agoraphobia) may lead to poor academic performance due to deficits in cognitive, motivational, and interpersonal functioning. Cognitively, GAD can produce task-irrelevant thoughts that have a deleterious effect on concentration (American Psychiatric Association, 2013a; Wine, 1971), which could impede task completion and, consequently, academic achievement. PTSD is characterized by traumatic event-induced distressing symptoms, including intrusive memory, persistent avoidance of trauma-related stimuli, maladaptive alterations in cognition and mood, and heightened physiological arousal and reactivity (American Psychiatric Association, 2013b). Research has shown that youth with PTSD are at high risk of attention, memory, executive function impairments (Barrera-Valencia et al., 2017), learning impairments (Samuelson et al., 2010), and social withdrawal (Elbert et al., 2009), which can impede school achievement.

Motivationally, anxious youth tend to have avoidance goals (Dickson & MacLeod, 2004), which may lead them to lower their achievement expectations as a coping mechanism (Abeler et al., 2011), and consequently reduce their effort on homework and exams. On the other hand, there may be an optimal level of anxiety that enhances task performance (Keeley et al., 2008; Owens et al., 2014).

Interpersonally, anxiety is associated with poor social skills, which interfere with peer interactions (Motoca et al., 2012) and may lead to low willingness to participate in group work or even go to school. Adolescents with social phobia have more trouble initiating and maintaining conversations (Mesa et al., 2014), are more likely to be deemed as unfriendly or lacking social skills (Inderbitzen-Nolan et al., 2007), and therefore may find it particularly stressful to speak in public, participate in group work, and ask for help from teachers, which collectively will make school less enjoyable. Individuals with panic disorder and agoraphobia have a strong tendency to avoid stress-evoking situations, which may impair social functioning and interfere with learning at school.

In general, longitudinal research supports a negative association between anxiety and educational outcomes (e.g., high school graduation), although the adverse effect of anxiety, compared with depression, tends to be smaller in magnitude and less consistent (for a meta-analysis, see Riglin et al., 2014). PTSD was prospectively associated with lower grade point average (GPA) and test scores among college students (Boyraz et al., 2016), but evidence for these associations was inconsistent for adolescents. Vilaplana-Pérez et al. (2020) found that prior lifetime diagnosis of PTSD was associated with lower odds of high school graduation and college attendance, whereas Scott et al. (2014) did not find a significant prospective association between PTSD symptoms and test scores 7 months later among fourth- to eighth-grade youth. Similarly, no consistent conclusion could be drawn from past research regarding how social phobia is associated with academic achievement. Some longitudinal studies found that social phobia predicted slow academic progress (Ranta et al., 2016) and lower teacher-reported grades (Leigh et al., 2021). By contrast, in other studies, social phobia did not predict GPA, school dropout, frequent class absences (Ranta et al., 2016), or exam scores (Xu et al., 2021). To our knowledge, no longitudinal studies have examined how panic disorder and agoraphobia are associated with school outcomes among adolescents.

Associations Between Externalizing Disorders and School Outcomes

The Externalizing Spectrum

Longitudinal studies have found that more externalizing symptoms predict lower subsequent academic achievement, including in samples of Mexican-origin youth (e.g., Roosa et al., 2012), with effect sizes as strong or even stronger than those for internalizing symptoms (e.g., Deighton et al., 2018; Zhang et al., 2019). Below, we review prior research examining prospective associations between specific externalizing disorders, including ADHD, ODD, CD, and school outcomes.[1]

Disruptive Behavior Disorders (ADHD, ODD, and CD)

ADHD, which involves deficits in attention and impulse control, is associated with lower verbal and working memory performance (Andersen et al., 2013; Calub et al., 2019) and impairments in executive attention (Hale et al., 2011), adaptive communication skills, and reading skills (Clark et al., 2002). Students with ADHD have trouble completing homework assignments (Langberg et al., 2016). Numerous two-wave longitudinal studies have shown that ADHD diagnosed in childhood is associated with a wide range of academic problems in adolescence and adulthood, although the effect size varies depending on the type of symptom (inattention vs. hyperactivity) and measure of achievement (Arnold et al., 2020; Lawrence et al., 2021).

ODD, a persistent behavioral pattern of defying authority, easily losing one's temper, and acting vindictively toward others, is associated with peer problems (Burke et al., 2010) and poor social functioning in school settings (Greene et al., 2002). ODD is prospectively associated with worse academic achievement among boys. Sayal et al. (2015) found that ODD at age 7 was associated with lower test scores at age 16 among boys, but not girls. Leadbeater and Ames (2017) found that ODD measured in adolescence predicted lower educational attainment in early adulthood and that increases in ODD symptoms throughout adolescence predicted lower educational attainment among boys but not girls.

CD includes externalizing symptoms such as theft, destruction of property, and physical assault. CD is associated with verbal deficits (Teichner & Golden, 2000) and affiliation with deviant peers (D. Chen et al., 2015), both of which seem likely to lead to poor educational outcomes. ADHD, ODD, and CD are associated with low social competence (Clark et al., 2002). All these factors could disrupt the learning process and lead to worse school outcomes. A review of longitudinal studies concluded that CD diagnosis was associated with lower educational attainment (Erskine et al., 2016).

[1] Although ADHD is categorized as a neurodevelopmental disorder (rather than an externalizing disorder) in the *Diagnostic and Statistical Manual of Mental Disorders*, fifth edition, it involves externalizing behaviors like hyperactivity, impulsivity, and, in some cases, aggression. These behaviors can lead to social, emotional, and academic problems. Therefore, we grouped ADHD with externalizing disorders.

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Substance Use Disorders

Substance use disorders (SUDs) are often included in the externalizing spectrum. Studies have shown that marijuana and alcohol use cause lasting impairment in neuropsychological and cognitive functioning among adolescents and young adults (e.g., Broyd et al., 2016; Thames et al., 2014), whereas others failed to find support for such associations (e.g., Schaefer et al., 2021). Instead, some studies suggest that substance use can lead to poor academic functioning primarily through affiliations with delinquent peers (Lynskey & Hall, 2000), socioemotional problems (Crosnoe et al., 2012), and absence from class (Chai et al., 2020). Moreover, numerous longitudinal studies have found that substance use problems in adolescence are prospectively associated with various indicators of academic underachievement, especially high school dropout (e.g., Busch et al., 2014; Meier, 2021).

Changes in Psychopathology and School Outcomes

Changes in psychopathology, denoted by the slope in a latent growth curve model, have different theoretical implications than the trajectory level. Although level is a static representation of the general level of psychopathology at a given point in time, the slope captures the collective consequences of ongoing transactions between the person and the environment and the upward or downward momentum of a person's increasing or decreasing levels of various forms of psychopathology. The reciprocal interaction between person and environment partially depends on initial psychopathology levels. Youth with early elevated levels of psychiatric disorders may be more vulnerable to environmental challenges. They may perceive these challenges as more uncontrollable compared with youth who started with lower levels of psychiatric disorders. However, slope effects differ from level effects in the sense that slopes capture reciprocal processes by incorporating the complexity of environmental challenges and contextual support. These person–environment interactions continue and accumulate over different time scales, ranging from momentary, moderate duration to the long term (Hopwood et al., 2022). Collectively, these transactions may contribute to differential academic outcomes over and above a youth's static level of psychopathology.

Notably, the reciprocal effects between person and environment are probabilistic and nonlinear. How individuals navigate the occurrence of multiple contextual factors over time is complex and difficult to model individually. Analogous to ideas from the dynamic systems perspective (e.g., Danvers et al., 2020), but spanning longer time spans, the slope characterizes how a person acts in and reacts to multilayered situations (and broader environmental contexts) and how these situations/contexts are, in turn, shaped by the individual's actions and reactions, creating a dynamic dependency between the two. Therefore, the slope becomes a unit of analysis that captures these ongoing transactional processes by accounting for unmeasured individual and contextual factors. These processes ultimately will serve to increase or decrease levels of psychopathology over time.

Thus, above and beyond initial or average levels of psychopathology, changes in psychopathology across adolescence may be especially relevant to later school outcomes. Youth experiencing rapid increases in psychiatric symptoms during middle and high school likely experience quite different person–environment transactions than those with steady or decreasing symptoms. For example, an adolescent who experiences an increase in anxiety and depression symptoms may have more difficulty coping with the many academic, social, and emotional challenges they face in the school context. Their disengagement and/or emotional outbursts in the classroom could evoke negative responses from teachers and peers that further increase their anxiety and depression symptoms, and they may seek out more dysfunctional environments (e.g., associating with deviant peers who are skipping school to engage in substance use). Together, these transactional processes can create a vicious cycle of progressively higher levels of anxiety and depression, leading to declining levels of academic functioning, which in turn contribute to even higher levels of anxiety and depression, and so on. Similarly, rising psychiatric symptoms may suggest exposure to uncontrollable, chronic environmental stressors such that the youth do not have sufficient time to recover from the stress (Sheth et al., 2017; Snyder et al., 2019). Additionally, these youth may lack social support to learn problem-solving skills and self-regulation strategies to develop adaptive response styles to stressors (Rueger et al., 2016). Without appropriate support from parents, teachers, and the community, these youth may react to, select, and evoke environmental changes in dysfunctional ways (Jackson & Warren, 2000), creating a vicious cycle over time.

Students from marginalized racial–ethnic groups may be particularly at risk for these vicious cycles given the structural and systemic challenges they face in their school and neighborhood environments. Specifically, they are more likely to come from lower socioeconomic backgrounds (KFF, n.d.; The Federal Reserve, 2023), attend underresourced K–12 schools (Baker et al., 2024), and live in high-poverty, high-crime neighborhoods (Browning et al., 2017). These students likely spend more time and energy overcoming poverty-related challenges, including but not limited to underage labor, sibling care responsibilities, and neighborhood harassment by gangs and delinquent individuals. Such chronic stressors exceed children's and adolescents' coping capacities and contribute to the development of psychiatric symptoms (Miller et al., 2021; Weinberger, 2023). Moreover, youth with limited time for learning (e.g., school absences) and inadequate support for social–emotional skills development often end up with poor academic performance (reaction effect; Lechner et al., 2019). Their poor academic performance frequently triggers negative feedback from teachers and peers (evocation effect) that subsequently leads to negative self-evaluations and emotions (Brummelman & Sedikides, 2023). Experiencing persistent frustration and reduced school belonging, these students may turn to substance use for stress relief (Anderson et al., 2024; Gaete et al., 2018). They may engage in disruptive behaviors and associate with delinquent peers to cope with identity confusion (selection effect; Demanet & Van Houtte, 2012; Mercer et al., 2017). These behaviors further reduce school attendance and learning engagement (Wood et al., 2012). Collectively, these patterns may manifest as increasing psychiatric symptoms or decreasing self-regulation, both associated with poor academic achievement.

Conversely, youth showing steady or slowly increasing psychiatric symptoms likely face less severe environmental challenges and may have more time to develop adaptive coping strategies and elicit support from parents, teachers, and the community. Consequently, these youth engage in adaptive or virtuous interactions with their environment. Consider students from marginalized ethnic groups who are more likely to experience poverty and discrimination. Those

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with supportive parents could focus more on learning rather than domestic responsibilities and consequently experience less conflict between home duties and schoolwork (reaction effect). Their academic progress may elicit additional teacher support (evocation effect; Košir & Tement, 2014; Zhou et al., 2024), which further enhances learning efficiency. The above example represents one of many possible person–environment interaction scenarios.

Our literature review indicates that numerous psychiatric disorders serve as risk factors for poor academic performance and negative school outcomes. Only a few studies have examined how changes in psychiatric disorders over time—at both the level of individual disorders and the broader internalizing and externalizing spectra—relate to educational outcomes. One study found that more rapid increases in internalizing symptoms were associated with slower increases in reading and math performance among older children (Keles et al., 2018), whereas two other studies (Grimm, 2007; Weidman et al., 2015) did not find these associations among adolescents. Studies on substance use showed more consistent results, that is, a more rapid increase in substance use across adolescence was associated with a more rapid decrease in academic grades (Henry, 2010) and a lower likelihood of attending and completing college education (King et al., 2006).

Given the lack of research on how changes in psychopathology are prospectively associated with educational outcomes independent of the level effect, further research is needed to investigate whether such associations exist and, if so, how the strength of the associations varies across different disorders and outcomes.

Psychopathology, Temperament, and School Outcomes

Psychopathology and temperament are conceptually and empirically related. Both constructs are prospectively associated with school outcomes. Given their conceptual overlap and similarity in predictive validity, a critical question emerges regarding the extent to which these constructs differ in predicting school outcomes. Temperament refers to individual differences in reactivity and self-regulation that emerge early in life and provide the foundation for adult personality traits (Rothbart, 2011). Temperament domains (i.e., effortful control, negative emotionality, and positive emotionality) and facets are prospectively associated with educational outcomes (for meta-analysis, see Nasvytienė & Lazdauskas, 2021). Temperament likely influences educational outcomes through cognitive, motivational, emotional, and interpersonal processes (Cheng et al., 2024). Using data from the same sample as the present study, Cheng et al. (2024) found that effortful control and its facets at age 10 predicted greater academic achievement, and negative emotionality at age 10 predicted worse academic achievement.[2] In addition, Cheng and colleagues found some support for the associations between changes in temperament and educational outcomes, but these slope associations were smaller in magnitude and less consistent than those between temperament levels and academic achievement.

Psychopathology and temperament could be considered sibling constructs for several reasons (Lawson & Robins, 2021). First, psychopathology and temperament share substantial overlap in their theorized nomological networks. Both our current literature review and that of Cheng et al. (2024) indicate that conceptually, similar mechanisms connect temperament and psychopathology to school outcomes, including cognitive processes, motivation, self-regulation, peer relationships, and student–teacher interactions. For example, both low effortful control and more externalizing symptoms can affect learning through disruptive classroom behavior, affiliation with deviant peers, and poor relationships with teachers.

Second, measures of psychopathology and temperament contain conceptually related items. Scales measuring internalizing disorders and negative emotionality both have items assessing anxiety and depression, and measures of externalizing disorders and effortful control both have items assessing impulsive and dysregulated behaviors. This observation applies to the scales used in the present study (see the Measures section) and other commonly used scales measuring internalizing disorders among adolescents (e.g., the Mood and Anxiety Symptom Questionnaire; Casillas & Clark, 2001) and externalizing disorders (e.g., the 2001 Child Behavior Checklist for Ages 6–18; Achenbach & Rescorla, 2001).

Third, not surprisingly, given the measurement overlap, prior research has documented empirical associations among (a) low effortful control and externalizing disorders, (b) high negative emotionality and both internalizing and externalizing disorders, and (c) low positive emotionality and depressive disorder (Kostyrka-Allchorne et al., 2020). Moreover, the surgency facet of positive emotionality includes pleasure derived from high-intensity or novel activities (Rothbart et al., 2004) and is conceptually associated with sensation seeking and impulsivity, which are linked to the development of externalizing symptoms (Byck et al., 2015; Martel et al., 2017). The affiliation facet of positive emotionality measures the desire for interpersonal warmth and connectedness (Rothbart et al., 2004). Although not conceptually related, affiliation is empirically associated with depressive symptoms, anxiety, and antisocial behaviors toward peers (Lawson & Robins, 2021; Snyder et al., 2015), perhaps because a desperate need for affiliation increases the risk of bullying and peer victimization among adolescents (Solomontos-Kountouri & Strohmeier, 2021).

Fourth, psychopathology and temperament show correlated developmental trajectories. For example, faster increases in effortful control are associated with faster decreases in externalizing symptoms (Atherton et al., 2019, 2020). Furthermore, psychopathology and personality are developmentally intertwined (Durbin & Hicks, 2014). Fifth, psychopathology and temperament share common biological underpinnings (Lemery-Chalfant et al., 2008; Tackett et al., 2013) and are influenced by similar environmental factors, including life events such as the death of a close family member (March-Llanes et al., 2017). Sixth, some researchers argue that psychopathology and temperament represent state and trait-level manifestations, respectively, of the same underlying processes (e.g., DeYoung et al., 2022). For example, individuals with temperamentally high levels of negative emotionality may be more inclined to experience episodes of severe depressed mood (i.e., MDD) and episodes of intense anxiety (i.e., GAD).

The above findings raise the question of whether any observed effects of psychiatric disorders on academic outcomes are driven by their overlap with temperament and, conversely, whether prior findings showing that temperament domains are associated with academic outcomes are driven by their overlap with psychiatric

[2] Using data from California Families Project, Cheng et al. (2024) examined the associations between temperament trajectories from ages 10 to 16 and school outcomes at age 17. The present study extends this research by incorporating data on psychiatric disorders from the same developmental period (ages 10–16) and new data on high school graduation and educational attainment at age 27.

disorders. Testing the incremental validity of psychopathology and temperament in predicting academic outcomes will help distinguish these two sets of sibling constructs and identify which construct(s) would produce stronger effects on school outcomes when targeted by interventions. This approach is also consistent with the movement toward integrating psychopathology and temperament/personality into a common framework (e.g., Wright & Hopwood, 2022), which replaces the conventional categorical psychiatric disorders with a set of continuous underlying dimensions specified by National Institute of Mental Health's Research Domain Criteria (Cuthbert, 2022).

The Present Study

The present study examined the association between trajectories of psychiatric disorders from late childhood through adolescence and school outcomes in late adolescence and early adulthood among a sample of Mexican-origin youth in the United States. Psychiatric disorders were assessed annually from age 10 to 16 (seven waves). Temperament was assessed at ages 10, 12, 14, and 16 (four waves). The present study addresses the following two research questions and hypotheses. Research Question 1: Do the level and slope of psychiatric disorders from age 10 to 16 predict worse school outcomes (high school GPA, achievement test scores, college entrance exam scores, high school course difficulty, high school graduation vs. dropout, and college attendance) in late adolescence and early adulthood? We predict that higher levels and more positive slopes (i.e., greater increases or smaller decreases) of the psychiatric disorder trajectories will be associated with worse school outcomes (Hypothesis 1a). Furthermore, we predict that these level and slope associations will hold for each psychiatric disorder, the internalizing spectra, the externalizing spectra, and the *p* factor (Hypothesis 1b). Within the internalizing spectrum, we predict that the effect sizes of MDD will be larger than those of anxiety-related disorders (Hypothesis 1c). In addition, we predict that the associations will hold for all school outcomes (Hypothesis 1d), with the most robust effects for educational attainment (i.e., high school graduation and college attendance) because there is more substantial evidence linking psychiatric disorders to educational attainment compared with other indicators of school success (Hypothesis 1e).

Research Question 2: Do psychiatric disorders predict school outcomes after controlling for temperament and vice versa (i.e., do psychiatric disorders and temperament have independent effects on school outcomes)? More specifically, do the levels (and slopes) of psychiatric disorders predict school outcomes after controlling for the levels (and slopes) of all temperament domains? Do the levels (and slopes) of temperament domains predict school outcomes after controlling for the levels (and slopes) of psychiatric disorders? These analyses are exploratory and will provide evidence for or against the incremental validity of psychopathology over temperament. Although we do not have predictions for Research Question 2, we expect that the temperament levels (and slopes) will predict school outcomes independent of psychiatric disorders, but not vice versa, because temperament may contribute to a broad spectrum of thoughts, feelings, and behaviors that improve academic performance, whereas psychopathology typically explains academic struggle. However, it is possible that the slopes of psychiatric disorders will uniquely predict school outcomes beyond temperament slopes, based on the conceptualization of psychopathology as maladaptive person–environment transactions (Hopwood et al., 2022), suggesting that psychopathology may be more distinguishable from temperament traits when explained from a transactional perspective (captured by slopes) rather than at a static time point.

The present study extends previous research in several ways. First, our longitudinal design follows the same participants from late childhood (age 10) through adolescence and into early adulthood (age 23), allowing us to examine the long-term consequences of psychopathology for academic success and failure and more precisely assess change over time in our predictors. Previous research has typically followed participants over a shorter time interval and/or included only two waves of data. Second, instead of focusing on a single educational outcome (e.g., high school graduation), we assessed several different key indicators of academic success, allowing us to evaluate whether the effects of psychiatric disorders vary across different components of academic success. Third, we assessed most school outcomes using data obtained from the California Department of Education, allowing us to avoid retrospective and social desirability biases in reports of these outcomes and reduce the risk that any observed associations with psychiatric disorders are due to shared method variance. This extends previous research that has relied primarily on self-, parent, and/or teacher reports of academic performance. Fourth, we examined whether psychiatric disorders predict academic outcomes after controlling for temperament differences (and vice versa); the findings from these analyses have important implications for theory and research on both psychopathology and temperament. Fifth, we examined these effects in a sample of Mexican-origin youth, which contributes to the generalizability of previous work in this area that has focused on predominantly European-background youth.

Method

Participants

This study used data from the California Families Project, an ongoing community-based longitudinal study of Mexican-origin youth and their parents (N = 674). Children were drawn at random from rosters of students from the Sacramento and Woodland, California, school districts. The focal child had to be in fifth grade, of Mexican origin, and living with his or her biological mother to participate in the study. Approximately 72.6% of the eligible families agreed to participate in the California Families Project, which was approved by University of California, Davis's Institutional Review Board (Protocol No. 217484-21). The children (50% female) were interviewed annually from fifth grade to 2 years posthigh school and then again at (approximately) 4, 6, 8, and 10 years posthigh school. The present study uses data from Waves 1–14 (collected in 2006–2025), when the children were in fifth grade (M_{age} = 10.8 years, SD = 0.50) to 10 years after high school (M_{age} = 27.5 years, SD = 0.62). Retention rates compared with the original sample are as follows: 84% (Wave 2), 86% (Wave 3), 88% (Wave 4), 90% (Wave 5), 88% (Wave 6), 89% (Wave 7), 89% (Wave 8), 87% (Wave 9), 87% (Wave 10), 80% (Wave 11), 81% (Wave 12), 75% (Wave 13), and 75% (Wave 14).

Participants were interviewed in their homes in Spanish or English, depending on their preference. Interviewers were all bilingual, and most were of Mexican heritage. Sixty-three percent of mothers and 65% of fathers had less than a high school education (median education level = ninth grade for both mothers and fathers); median total household income was between $30,000 and $35,000.

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With regard to generational status, 84% of mothers and 89% of fathers were first-generation immigrants. Eighty-one percent of students (N = 549) resided in two-parent families, and 19% of students (N = 125) resided in single-parent families. In the present study, we used data for all available participants (i.e., no exclusions were applied) and reported all analyses conducted to address our research questions.

Transparency and Openness

The preregistration, deviations from the preregistration, data analysis script, and additional online material are available on the Open Science Framework at https://osf.io/8ydqt. We are not legally or ethically allowed to post data for this project because the participants in the study have not given informed consent to have their data publicly shared, and we do not have institutional review board approval to post data publicly.

Missing Data

We compared individuals who did versus did not have school outcome data on the main study variables (i.e., psychiatric disorders, gender, nativity) measured at Wave 1. Results are reported in Supplemental Table S1. With a few exceptions, psychiatric disorders at age 10 were not associated with having versus not having data for high school GPA, achievement test scores, high school course difficulty, and college attendance. However, participants reporting more externalizing symptoms were more likely to have data for high school graduation and college entrance exam scores compared with those reporting fewer externalizing symptoms. U.S.-born participants were more likely than Mexico-born participants to have data for all school outcomes except college attendance, reflecting the fact that Mexico-born participants were more likely to move back to Mexico and consequently no longer attend school in California.

Measures

Psychiatric Disorders

The National Institute of Mental Health Diagnostic Interview Schedule for Children–IV (DISC-IV; Shaffer et al., 2000) was administered to the child annually from age 10 (fifth grade) to 17 (12th grade). The present study focuses on psychiatric data from age 10 to 16 to parallel the time frame for the temperament assessments. The DISC-IV is a comprehensive psychiatric interview that assesses mental health disorders using *Diagnostic and Statistical Manual of Mental Disorders*, fourth edition, criteria; it is the most widely used mental health interview that has been tested in clinical and community populations and validated in both English and Spanish (Costello et al., 1984). Responses were recorded dichotomously (0 = *no*, 1 = *yes*) as the symptom being present or not in the past year. Total symptom counts will be created for each psychiatric disorder at each wave.

Internalizing Disorders. The MDD module includes questions about depressed mood, diminished interest or pleasure in activities, and physical symptoms (e.g., sleeping more than usual). The GAD module includes questions about general worry and concern (e.g., tenseness) and physical symptoms (e.g., frequent stomachaches). The social phobia module includes questions about embarrassment or humiliation caused by social situations. The panic disorder module includes questions about the occurrence of intense fear and discomfort and the associated somatic and cognitive symptoms. The agoraphobia module includes questions about anxiety caused by places or situations from which escape might be difficult or embarrassing. The PTSD module includes questions about irritability after the trauma experience or reexperience of the traumatic event. The agoraphobia, panic disorder, and social phobia modules were only administered at ages 15 and 16; consequently, we averaged across the symptom counts at ages 15 and 16 instead of estimating growth curve models of changes in symptom counts. Internalizing symptoms were operationalized as the total possible symptom counts of MDD, GAD, social phobia, panic disorder, agoraphobia, and PTSD.

Externalizing Disorders. The ADHD module includes questions about attention-related behaviors such as "[Did you have] trouble keeping your mind on task for more than a short period of time?" and hyperactivity problems such as "[Did you] often climb on things/run around when you weren't supposed to?" The ODD module includes questions about irritability such as "[Did you] lose your temper?" and defiant behaviors such as "[Did you] do things on purpose that caretakers said not to do?" The CD module includes questions about more severe delinquent behavior, such as "[Did you] break/damage someone else's things on purpose?" and "[Did you] bully someone smaller who wouldn't fight back?" The SUD module includes questions about the use of alcohol, marijuana, nicotine, and several other illicit substances (amphetamines, lysergic acid diethylamide, etc.). For each substance, questions are asked about frequency of use in the past year, problems associated with use (e.g., missing school or work), and dependence (e.g., withdrawal). Externalizing symptoms were operationalized as the total symptom counts of ADHD, ODD, CD, and SUDs.

Total Symptoms. Total symptoms (i.e., the p factor of psychopathology) were operationalized by summing all internalizing and externalizing symptoms at each wave.

Temperament

Temperament was assessed at ages 10, 12, 14, and 16 via self- and mother reports using the short version of the Early Adolescent Temperament Questionnaire–Revised (EATQ; Ellis & Rothbart, 2001; for items used in EATQ, see Putnam, n.d). Ratings for each item were made on a 4-point scale ranging from 1 (*not at all true of you/your child*) to 4 (*very true of you/your child*). The EATQ assesses three broad temperament domains: effortful control, negative emotionality, and positive emotionality. For each temperament domain, we created latent variables using parcels that combined self- and mother reports.

The effortful control scale (16 items) assesses inhibitory control (the ability to plan and suppress inappropriate impulses), activation control (the ability to perform an action or pursue goals when there are competing desires), and attention control (the ability to focus and shift attention when needed).

The negative emotionality scale (19 items) assesses fear (unpleasant affect derived from anticipating distress), frustration (negative affect related to ongoing tasks being interrupted), depressed mood (unpleasant affect, lowered mood, and lack of enjoyment in activities), aggression (hostile reactivity to negative stimuli including person- and object-directed

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violence), and shyness (behavioral avoidance of novelty and social challenges).

The positive emotionality scale (19 items) assesses surgency (pleasure derived from activities involving high intensity or novelty) and affiliation (the desire for warmth and closeness with others). However, surgency (14 items) and affiliation (nine items) do not form a coherent superordinate factor (Lawson et al., 2021; Snyder et al., 2015), so we examined these scales separately instead of computing a positive emotionality domain score.

Academic Outcomes

The following academic outcomes were obtained from the California Department of Education and local school districts in Sacramento and Woodland counties: high school GPA, achievement test scores, college entrance exam scores, high school graduation (vs. dropout), and high school course difficulty (i.e., number of advanced placement [AP] courses taken in high school). When data were missing (e.g., when a child moved to another state or Mexico during their school-aged years), we used data from self- and parent reports (if available) to maximize the sample size.

High School GPA. High school GPA was calculated at the end of high school (i.e., end of 12th grade). If the student dropped out of high school, we used the student's cumulative GPA at the year of withdrawal. Following standard practice in California high schools, AP courses were weighted 1 point higher than non-AP courses (e.g., an "A" = 5.0) given the difficulty of getting good grades in these more advanced courses. The resulting variable had 650 cases (M = 2.58, SD = 0.80, min = 0.21, max = 4.6); 11.2% of the cases are based on self- or parent-reported high school GPA.

Achievement Test Scores. We created a composite measure of achievement test scores based on the mean of standardized scores on the California Standards Test from Grades 9 to 12 and the California High School Exit Examination, which was required for all California high school students (N = 587, M = 0.00, SD = 1.00). All achievement test scores come from school records.

College Entrance Exam Scores. We created a composite measure based on the mean of standardized scores on the Scholastic Aptitude Test and the American College Testing (N = 250, M = 0.00, SD = 1.00). Overall, 32.8% of the cases were based on self-reported test scores (all were Scholastic Aptitude Test scores). Most participants did not take the Scholastic Aptitude Test or the American College Testing during high school.

High School Course Difficulty. Course difficulty reflects the number of AP courses that a student took in high school (N = 542, M = 1.22, SD = 2.47, min = 0, max = 16). One hundred eighty participants (33.2%) took at least one AP course (M in this subsample = 3.66, SD = 3.07); all the data on AP courses come from school records.

High School Graduation (vs. Dropout). We created a dichotomous variable indicating whether each participant graduated (coded as 1; n = 569) or dropped out (coded as 0; n = 98) of high school (dropout rate = 15%); 10% of the cases are based on self- or parent reports of high school graduation. Participants who received a high school diploma or general education diploma were counted as high school graduates.

College Attendance. We created a dichotomous variable indicating whether each participant attended (coded as 1; n = 365) or did not attend (coded as 0; n = 302) college (college attendance rate = 55%). Participants attending either 4-year colleges/universities or community colleges were counted as attending college; participants attending vocational/technical schools were not counted as attending college. All cases are based on self- or parent reports of school attendance.

Nativity

We created a dichotomous nativity variable comparing youth born in Mexico (29%; coded as 0) to youth born in the United States (71%; coded as 1).

Data Analysis

Data cleaning and analyses were conducted in R (R Core Team, 2022) via R Studio Version 2023.06.2 and Mplus Version 7.31 (Muthén & Muthén, 1998/2017). We used the following R packages to analyze the data and visualize the results: psych (Revelle, 2022), lavaan (Rosseel, 2012), ggplot2 (Wickham, 2016), and MplusAutomation (Hallquist & Wiley, 2018). We used a robust maximum likelihood estimator to account for nonnormal distributions of observed variables and full information maximum likelihood procedure to account for missing data (Allison, 2003; Schafer & Graham, 2002). All models predicting continuous outcomes with growth curves were conducted in R. All models predicting categorical outcomes (high school graduation, college attendance) with growth curves and multigroup analyses were conducted in Mplus.

Hypotheses 1a–1d of Research Question 1 were tested using univariate latent growth curve models. To facilitate interpretation, we specified a linear growth trajectory for all psychiatric disorders by fixing the slope factor loadings to −0.5, −0.333, −0.167, 0, 0.167, 0.333, and 0.5. This way, the level was centered at the midpoint of the assessment waves (age 13), which best represents the average level of psychiatric disorder over the entire assessment period. A one-unit increase in the slope factor reflects a change in psychiatric symptoms across 7 years (ages 10–16). In each model, we examined how the level and slope of one psychiatric disorder were associated with one indicator of academic success. For these analyses, we used an α level of .05 (two-tailed) as our inference criteria and interpreted p values below this threshold to be significant. We did not adjust the α level because no universal null hypothesis was made, all hypotheses were preestablished, analyzed outcomes were correlated but measured different aspects of academic achievement, and predictor measures were previously published and validated (Armstrong, 2014; Rothman, 1990; Streiner & Norman, 2011). We reported exact p values in the Supplemental Material for all tests.

To determine whether the findings of Research Question 1 generalize across sex and nativity, we used multigroup structural equation models to test for sex and nativity differences in the associations between psychiatric disorder trajectories and school outcomes. We compared the fit of models in which the associations were constrained to be equal across groups to models in which these parameters were freely estimated. If the constrained model did not fit significantly worse than the unconstrained model, then we concluded that the associations between psychiatric disorders and school outcomes were similar for boys and girls and youth born in the United States and Mexico. Group differences were tested separately for the level and slope effects. We interpret differences in model fit based on significant results of chi-square test, change in

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comparative fit index larger than .01, change in root-mean-square error of approximation larger than .015, and change in Bayesian information criterion larger than 6 (F. F. Chen, 2007; Cheung & Rensvold, 2002; Kass & Raftery, 1995; Meade et al., 2006).

Research Question 2 (incremental validity) was tested using multiple regression analyses. We only examined the incremental validity of psychiatric disorders that had significant associations with school outcomes. Before testing incremental validity, the level and slope factor scores were saved from the linear growth models of each psychiatric disorder and each temperament domain. When testing the incremental validity of the level of a particular psychiatric disorder, we regressed one school outcome on this psychiatric disorder level while controlling for all temperament domain levels. When testing the incremental validity of the slope of a particular psychiatric disorder, we regressed one school outcome on this psychiatric disorder slope while controlling for its level and all temperament domain slopes.

Similarly, we only tested the incremental validity of temperament domains that had significant associations ($p < .05$) with school outcomes. To determine which temperament domain-school outcome association is statistically significant, we reran all the growth curve models in Cheng et al.'s (2024) study by fixing the levels at the midpoint of the assessment, the same as was done for psychiatric disorder levels. We then followed the same procedures described above to test whether the temperament effects held after controlling for internalizing and externalizing levels and slopes.

Given that all analyses related to Research Question 2 were exploratory, we used an α level of .05 (two-tailed). No adjustment to α levels was made to maintain the power of detecting the effect and avoid Type II error.

Results

Descriptive statistics for psychiatric disorders and temperament domains at each measurement occasion are presented in Supplemental Table S2. Correlations between psychiatric disorders and school outcomes are presented in Supplemental Tables S3–S13. Overall, psychiatric disorders at each wave had small-to-medium (Funder & Ozer, 2019) negative associations with school outcomes. For MDD and GAD, their associations with school outcomes weaken at later assessment waves. For ADHD, ODD, internalizing symptoms, and total symptoms, the correlations shift from negative at earlier assessment waves to slightly positive at later waves. Linear growth curve statistics for psychiatric disorders are presented in Supplemental Table S14. Psychiatric disorder trajectories of 50 randomly selected participants were visualized in Supplemental Figure S1.

Research Question 1: Do the Level and Slope of Psychiatric Disorders From Age 10 to 16 Predict Worse School Outcomes in Late Adolescence and Early Adulthood?

Level Effects

Consistent with Hypothesis 1a, psychiatric disorder levels predicted worse academic outcomes in high school (see Table 1; for p values, confidence intervals, and other detailed information, see Supplemental Tables S15 and S16), including lower GPA, achievement test scores, college entrance exam scores, course difficulty, likelihood of high school graduation, and likelihood of college attendance (β ranged from −.07 to −.48). The only exception is that panic disorder symptoms were associated with higher achievement test scores. However, examination of the bivariate scatterplot showed that this association was driven by four outliers, and excluding these outliers resulted in a nonsignificant association between panic disorder and achievement test scores. Consequently, we do not consider this a robust finding.

In contrast to Hypothesis 1b, particular psychiatric disorder levels had no association with most school outcomes. These psychiatric disorders include GAD, social phobia, agoraphobia, PTSD, and ODD. Partially consistent with Hypothesis 1c, compared with levels of anxiety-related disorders, MDD levels had larger effects in predicting high school GPA and college attendance. However, they showed a similar magnitude of effects in predicting other school outcomes. In contrast to Hypothesis 1d, we found no statistically significant associations between most psychiatric disorder levels and particular school outcomes, including college entrance exam scores and high school graduation. In contrast to Hypothesis 1e, the associations between psychiatric disorder levels and college attendance were no more robust than the associations between psychiatric disorder levels and other school outcomes. Psychiatric disorder levels had equally robust associations with high school GPA, college attendance, and course difficulty.

Slope Effects

Contrary to Hypothesis 1a, the psychiatric disorder slope effects were in the opposite direction from the level effects. That is, faster increases in psychiatric symptoms from age 10 to 16 predicted better school achievement after controlling for the corresponding level effects. We will return to these seemingly paradoxical findings in the Sensitivity Analysis section.

In contrast to Hypothesis 1b, particular psychiatric disorder slopes had no association with school outcomes. These psychiatric disorders include PTSD, ODD, and SUDs.

Contrary to Hypothesis 1c, MDD slopes had smaller effects than GAD slopes.

Partially consistent with Hypothesis 1d, GAD slopes were positively associated with school outcomes. However, the slopes of most other psychiatric disorders were not associated with college entrance exam scores or high school course difficulty.

In contrast to Hypothesis 1e, the associations between psychiatric disorder slopes and educational attainment were no more robust than the associations between psychiatric disorder slopes and other school outcomes.

Generalizability Across Gender and Nativity Groups

Results of multigroup analyses are presented in Supplemental Tables S17 and S18. For all models, constraining the parameters to be equal across gender and nativity did not lead to a significant reduction in fit. None of the comparisons were significant for more than one of the preregistered criteria. For models that required comparing Bayesian information criterion scores, the freely estimated model tended to have higher Bayesian information criterion scores (worse fit) than the constrained model. The only exceptions were models of SUDs predicting college attendance, for which we found strong evidence that the unconstrained slope and unconstrained level models fit better than the constrained model. However,

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Table 1

Main Effects and Incremental Validity of Psychiatric Disorder Trajectories (Levels and Slopes) Over Temperament Trajectories (Levels and Slopes) in Predicting School Outcomes

Predictor	High school GPA	Achievement test score	College entrance exam score	High school course difficulty	High school graduation	College attendance
Total symptoms						
Level	**−.18** (.04)	**−.10** (**.21**)	−.10 (—)	**−.12** (.02)	−.06 (—)	**−.18** (.05)
Slope	**.15** (**.14**)	**.28** (**.19**)	.17 (—)	−.09 (—)	**.24** (**.35**)	**.17** (**.30**)
Internalizing symptoms						
Level	**−.23** (.04)	**−.21** (**.17**)	−.14 (—)	**−.12** (.04)	−.05 (—)	**−.25** (.04)
Slope	**.29** (**.20**)	**.39** (**.24**)	.23 (—)	.17 (—)	.26 (—)	**.30** (**.47**)
MDD						
Level	**−.13** (.03)	−.05 (—)	−.05 (—)	**−.07** (.03)	.06 (—)	**−.14** (.02)
Slope	**.20** (**.16**)	**.26** (**.16**)	.15 (—)	.07 (—)	.20 (—)	**.17** (**.26**)
GAD						
Level	−.09 (—)	−.09 (—)	−.08 (—)	−.06 (—)	.09 (—)	−.10 (—)
Slope	**.30** (**.16**)	**.39** (**.21**)	**.32** (**.16**)	**.22** (**.13**)	.22 (—)	**.37** (**.42**)
Social phobia	.00 (—)	.03 (—)	.01 (—)	.01 (—)	.10 (—)	.08 (—)
Panic disorder	.03 (—)	**.13** (**.18**)	−.06 (—)	.03 (—)	.18 (—)	.08 (—)
Agoraphobia	−.03 (—)	.03 (—)	−.04 (—)	−.03 (—)	.28 (—)	.09 (—)
PTSD						
Level	−.11 (—)	−.05 (—)	−.05 (—)	−.02 (—)	−.10 (—)	**−.14** (−.06)
Slope	.03 (—)	.04 (—)	.01 (—)	.02 (—)	.02 (—)	−.03 (—)
Externalizing symptoms						
Level	**−.17** (.02)	−.04 (—)	−.05 (—)	**−.11** (.00)	−.12 (—)	**−.13** (.03)
Slope	.05 (—)	**.14** (.08)	.04 (—)	−.02 (—)	.22 (—)	.00 (—)
ADHD						
Level	**−.18** (.03)	−.03 (—)	−.09 (—)	**−.15** (.00)	−.02 (—)	**−.19** (−.03)
Slope	.11 (—)	**.24** (**.15**)	.19 (—)	.09 (—)	**.28** (**.37**)	**.17** (**.25**)
ODD						
Level	−.06 (—)	.06 (—)	.03 (—)	−.04 (—)	.06 (—)	−.02 (—)
Slope	.09 (—)	.09 (—)	.05 (—)	.01 (—)	.22 (—)	−.02 (—)
CD						
Level	**−.21** (−.05)	**−.09** (.07)	−.01 (—)	**−.09** (−.03)	**−.16** (−.05)	**−.14** (−.17)
Slope	.03 (—)	.09 (—)	−.19 (—)	**−.08** (−.07)	−.10 (—)	**−.16** (−.17)
SUDs[a]						
Level	**−.35** (**−.09**)	**−.48** (−.02)	.68[b] (—)	**−.21** (−.07)	−1.10[b] (—)	−.47 (—)
Slope	.14 (—)	.38 (—)	−.90[b] (—)	.05 (—)	.95[b] (—)	.34 (—)

Note. Values on the left are standardized regression coefficients, except for the values for high school graduation and college attendance, which are standardized log odds. Values in parentheses are standardized regression coefficients or standardized log odds after controlling for the four temperament domain levels and slopes (effortful control, negative emotionality, surgency, affiliation). Dashes on parentheses indicate that the incremental effect was not tested due to a nonsignificant main effect. Effects significant at $p < .05$ are presented in bold. GPA = grade point average; MDD = major depressive disorder; GAD = generalized anxiety disorder; PTSD = posttraumatic stress disorder; ADHD = attention-deficit/hyperactivity disorder; ODD = oppositional defiant disorder; CD = conduct disorder; SUDs = substance use disorders.

[a] SUD trajectories were modeled from age 11 to 16 to solve model nonconvergence issues. [b] The large but nonsignificant effect sizes are likely statistical artifacts due to the zero-inflated distribution of SUD (especially at earlier ages) combined with the dichotomous scaling of high school graduation and college attendance.

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the effects of the SUD levels and slopes were not statistically significant in any of these models. Overall, these results suggest that findings were similar for boys and girls and for youth born in Mexico and in the United States.

Research Question 2: Do Psychiatric Disorders Predict School Outcomes After Controlling for Temperament and Vice Versa?

Table 1 presents the results of analyses examining the incremental validity of psychiatric disorders over temperament (for regression coefficients, see Supplemental Table S19). In most cases, psychiatric disorder levels did not predict unique variances in school outcomes, and their level effects decreased substantially after controlling for the four temperament domains (effortful control, negative emotionality, surgency, and affiliation). However, a few notable exceptions emerged: First, SUDs predicted unique variances in high school GPA after controlling for temperament ($\beta = -.09$). Second, in particular cases, controlling for temperament led to a change in the direction of the effect, a pattern referred to as negative suppression (Paulhus et al., 2004). Specifically, the effect of total symptoms level on achievement test scores changed from −.10 to .21, and the effect of internalizing symptoms level on achievement test scores changed from −.21 to .17.

In most cases, psychiatric disorder slopes predicted unique variance in school outcomes after controlling for the temperament slopes, with the effect sizes increasing in about half the cases, suggesting a more typical suppression effect.

Table 2 presents the results of analyses examining the incremental validity of temperament over psychiatric disorders (for regression coefficients, see Supplemental Table S20). The temperament levels consistently predicted school outcomes over and above psychiatric

Table 2
Main Effects and Incremental Validity of Temperament Trajectories Over Psychiatric Disorders in Predicting School Outcomes

Predictor	High school GPA	Achievement test score	College entrance exam scores	High school course difficulty	High school graduation	College attendance
Effortful control						
Level	**.42** (**.44**)	**.41** (**.46**)	**.18** (**.16**)	**.30** (**.30**)	**.44** (**.86**)	**.36** (**.59**)
Slope	.05 (—)	−.05 (—)	**−.16** (**−.16**)	−.04 (—)	.03 (—)	.09 (—)
Negative emotionality						
Level	**−.15** (**−.12**)	**−.19** (**−.22**)	−.10 (—)	**−.13** (**−.12**)	**−.20** (**−.49**)	**−.25** (**−.35**)
Slope	**.10** (**.09**)	**.19** (**.14**)	**.22** (**.21**)	**.09** (.08)	.04 (—)	.01 (—)
Surgency						
Level	−.02 (—)	.02 (—)	**.18** (**.17**)	.03 (—)	.03 (—)	−.02 (—)
Slope	.05 (—)	**.11** (**.10**)	−.08 (—)	.02 (—)	**.27** (**.34**)	.00 (—)
Affiliation						
Level	**.14** (**.14**)	.00 (—)	−.02 (—)	**.13** (**.09**)	**.18** (**.30**)	.11 (—)
Slope	−.01 (—)	**.11** (**.11**)	.06 (—)	**−.11** (**−.11**)	.15 (—)	.15 (—)

Note. Values outside of the parentheses are standardized regression coefficients (high school GPA, achievement test scores, college entrance exams, course difficulty) or standardized log odds (high school graduation and college attendance). Values in parentheses are standardized regression coefficients or standardized log odds after controlling for internalizing and externalizing levels and slopes. Dashes on parentheses indicate that the incremental effect was not tested due to a nonsignificant main effect. Effects significant at $p < .05$ are presented in bold. GPA = grade point average.

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disorder levels (i.e., after controlling for internalizing and externalizing symptoms). The effect sizes of temperament levels barely changed after controlling for psychiatric disorder levels and in all cases remained statistically significant. In cases wherein the target outcomes are high school graduation and college attendance, temperament level effects became larger after controlling for psychiatric disorder levels, suggesting a suppression effect. Similarly, the temperament slopes continued to predict unique variance in school outcomes after controlling for the psychiatric disorder slopes.

Sensitivity Analysis

We conducted a set of non-pre-registered exploratory analyses to test the robustness of the counterintuitive slope effects in Research Question 1. Together, these follow-up analyses have convinced us that the effects are robust, at least for the internalizing symptoms in the present study.

First, we reparametrized the latent growth curve models in several different ways: (a) The intercept was fixed at the baseline by setting slope factor loadings to 0, 0.166, 0.333, 0.5, 0.666, 0.833, and 1; (b) the intercept was fixed at the end line by setting slope factor loadings to −1, −0.833, −0.666, −0.5, −0.333, −0.166, and 0; (c) each school outcome was regressed on the slope, but not the level, but the slope and level were allowed to covary; (d) each school outcome was regressed on both level and slope, but the covariance of the level and slope was fixed to zero. None of these reparameterization attempts, which were implemented independently, changed the direction of the slope effects, and most effects remained significant. (e) Finally, we modeled curvilinear slopes for several psychiatric disorders that showed unexpected slope effects (i.e., GAD, ADHD, internalizing symptoms, and externalizing symptoms) and examined their associations with school outcomes (Supplemental Table S21). The results show that the direction and magnitude of the curvilinear slope effects closely resemble the linear slope effects.

Second, we examined the test–retest correlations of internalizing symptoms across waves. As expected, internalizing symptoms were strongly positively correlated (Supplemental Table S10), ruling out the possibility that mistakes were made during the computation of internalizing symptoms. We found the same pattern for externalizing symptoms (Supplemental Table S11).

Third, we removed multivariate outliers. Fourth, when regressing each school outcome on the level and slope, we controlled for demographic variables including child gender, nativity, household income, and parental education of the child at age 10. Neither attempt systematically changed the direction of the slope effects.

Fifth, we conducted subgroup analyses to test whether the internalizing and externalizing slope effects were positive among high-, medium-, and low-achieving adolescents and confirmed that this was the case.

Sixth, we examined the correlations between internalizing symptoms at each wave and school outcomes. We found that the association between internalizing symptoms and achievement became less negative, and in some cases positive, as the temporal distance of the measurement decreased (Supplemental Table S10). For MDD and GAD in particular, their associations with school outcomes changed from negative to positive as these disorders were measured closer to the end of high school (Supplemental Tables S3 and S4). When examining the correlations between externalizing symptoms at each wave and school outcomes, we again found the same pattern (Supplemental Table S11).

Last, the temperament domain of negative emotionality provides a conceptual replication of the internalizing symptom slope effects. Specifically, the negative emotionality and internalizing symptom slopes are positively associated, supporting their conceptual and empirical link (Supplemental Table S12), and the negative emotionality slope is also positively associated with school success (Supplemental Table S22). To examine whether the externalizing slope effects are measure-specific, we examined the slope effects for two alternative measures of externalizing symptoms as a conceptual replication: an adolescent delinquency scale from the National Longitudinal Study of Adolescent to Adult Health and a substance use scale adapted from Elliott et al. (1985). In contrast to the positive slope effects for the DISC-IV measure of externalizing symptoms, we found negative slope effects for the two alternative measures of delinquency and substance use (see Supplemental Table S23), consistent with prior research and theory.

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Discussion

The present study used multimethod data from a longitudinal study of 674 Mexican-origin youth to investigate the link between a wide range of psychiatric disorders and multiple academic outcomes assessed via school records. Our hypotheses for Research Question 1 were partially supported. First, consistent with our hypotheses, higher levels of psychiatric disorders predicted worse school outcomes, particularly high school GPA, achievement test scores, high school course difficulty, and college attendance. The effect sizes ranged from small to moderate (Funder & Ozer, 2019). Second, contrary to our expectation, faster increases (or slower decreases) in psychiatric disorders from age 10 to 16 predicted higher achievement after controlling for psychiatric disorder levels. In particular, faster increases in internalizing symptoms were positively associated with school outcomes. Exploratory analyses showed that the level and slope effects were similar for boys and girls and for youth born in the United States and in Mexico.

Exploratory analyses for Research Question 2 showed that, in most cases, psychiatric disorder levels did not predict unique variance in school outcomes after controlling for temperament levels, whereas the psychiatric disorder slopes predicted school outcomes after controlling for temperament slopes. Conversely, temperament levels and slopes generally predicted school outcomes after controlling for psychiatric disorder levels and slopes, suggesting that temperament is a more robust predictor of school outcomes than psychiatric disorders. Importantly, our results do not imply that psychiatric disorders are irrelevant to learning or predicting school outcomes but rather that psychiatric disorders lose their predictive power because they are partially redundant with temperament. The part that is redundant is essentially the part that predicts achievement. More broadly, the present findings support the movement toward conceptualizing psychiatric disorders in terms of a small number of continuous underlying dimensions—dimensions that, at least in part, map onto the primary temperament domains.

Psychiatric Disorder Levels and School Outcomes

Internalizing Symptom Level Effects

Students with more internalizing symptoms at age 13 had lower GPAs and achievement test scores, took fewer challenging courses in high school, and were less likely to attend college compared with students with fewer internalizing symptoms. These effects were mainly driven by MDD. None of the anxiety disorders was significantly associated with school achievement, although GAD was negatively associated with five out of the six outcomes (panic disorder, as noted earlier, was associated with achievement test scores, but this effect was driven by a few outliers). These findings align with previous meta-analyses (e.g., Wickersham et al., 2021) showing a robust association between depression and academic achievement and studies showing inconsistent evidence for the association between anxiety disorders and achievement (e.g., for anxiety, see Riglin et al., 2014; for PTSD, see Vilaplana-Pérez et al., 2020, and Scott et al., 2014; for social phobia, see Leigh et al., 2021, and Xu et al., 2021).

Although not directly tested, our findings are consistent with the hypothesized mediating processes, whereby students with more depressive symptoms (compared with those with fewer symptoms) exhibit deficits in cognitive, motivational, and interpersonal functioning that impede learning and performance in school. Specifically, students prone to depressive affect have diminished executive function capacities (i.e., problems with attention, inhibition, and working memory), weaker achievement motivation, and interpersonal difficulties that might interfere with the development of supportive relationships. More concretely, they will have difficulty paying attention in class, put less effort into homework, and have worse relationships with their teachers and peers, which will collectively contribute to poor academic performance and disengagement from the school context. Thus, we have evidence that depression is associated with poor school outcomes and that depression is associated with many risk factors for poor school performance (e.g., poor executive function, low achievement motivation, dysfunctional relationships). However, there is limited research testing these specific mediation pathways. Future research should seek to identify the most robust and potent mediators of the association between depression and school outcomes, which will help build more nuanced theoretical models and also identify viable targets of interventions aimed at reducing the negative impact of depressive symptoms on achievement.

It is worth noting that neither internalizing symptoms nor any of the individual internalizing disorders were significantly associated with two school outcomes: college entrance exam scores and high school graduation. The association between internalizing symptoms and college entrance exams ($\beta = -.14$) would have been statistically significant in the full sample, but only about one third of participants ($n = 250$) took these exams; this subset of students likely includes the most academically successful students in the sample, which would truncate the distribution of test scores, making it even more difficult to find effects. Moreover, students who took the college entrance exams, compared with those who did not, were more likely to have parents with a higher education level and more likely to have attended better resourced schools in less violent neighborhoods. These contextual factors may buffer students from the negative impact of psychiatric disorders on school outcomes. In other words, students might be able to compensate for the detrimental effect of psychiatric disorders by taking advantage of family, school, and neighborhood resources and support. Future studies should examine whether these and other contextual factors moderate the association between internalizing symptoms and school outcomes to better understand the sources of resilience in the youth's environment.

The association between internalizing symptoms and graduating from high school was also in the predicted direction ($\beta = -.09$), but not statistically significant. Although a wide range of factors influence school outcomes examined in the present study, high school graduation is arguably the most multiply determined. In fact, of the students in the present sample who dropped out of high school, only about one fourth cited academic reasons for doing so; the rest cited nonacademic reasons such as needing to get a job, getting pregnant, and responsibilities at home. Another reason for the nonsignificant effect of internalizing symptoms is that most youth in the present study experienced relatively few internalizing

symptoms ($M = 6.3$ symptoms; 39 participants had zero symptoms, Supplemental Table S2), which might have adversely affected academic performance, but not to the extent that it led to school dropout. Finally, most parents in the present study have high educational expectations for their children; on average, these parents expected their children to obtain at least a master's degree (Aceves et al., 2020). These parents are more inclined to support their children's learning and closely monitor their children's daily activities and whereabouts, which could potentially mitigate the adverse impact of internalizing symptoms on high school graduation.

Externalizing Symptoms Level Effects

Students with more externalizing symptoms had lower GPAs, took fewer challenging courses in high school, and were less likely to attend college compared with those with fewer externalizing symptoms. Externalizing symptoms had small to moderate effects, and these effects were mainly driven by ADHD, CD, and SUDs. Our findings align with prior research showing that ADHD and CD are more robust predictors of academic achievement than ODD (Galéra et al., 2009). Moreover, the moderate to large effect sizes observed for SUDs in our study align with the reports from a prior meta-analysis (Jeynes, 2022).

Our findings are consistent with the hypothesized mediating processes, whereby externalizing symptoms contribute to deficits in cognitive, behavioral, and interpersonal functioning that impede learning and academic performance. Specifically, students with more ADHD and CD symptoms exhibit disrupted attention control, reduced persistence in learning, diminished effort on homework, and more disruptive behaviors in the classroom. Students exhibiting cognitive and behavioral dysfunctions face more challenges in maintaining high GPAs, taking more advanced coursework, and attending college that are more demanding of sustained attention, persistence, and self-regulation. Additionally, students with more CD symptoms have poor interpersonal skills that undermine student–peer and student–teacher relationships. Lacking peer and teacher support at school threatens students' sense of belonging, which in turn impedes achievement motivation and learning-approach goals. Students with more CD and SUD symptoms may affiliate with deviant peers and become involved in gangs, impeding time spent on academic pursuits.

Notably, despite the higher level and greater variance in ADHD symptoms ($M = 2.18$, $SD = 2.6$) compared with CD ($M = 0.69$, $SD = 1.26$) in the current sample, CD exerted a more pronounced adverse effect than ADHD on standardized test scores. This finding is not consistent with prior research suggesting that ADHD is prospectively associated with lower achievement test scores (Arnold et al., 2020). This discrepancy may be due to our focus on a community sample with relatively low levels of ADHD, in contrast to prior research that compared students who were or were not diagnosed with ADHD.

Finally, neither externalizing symptoms nor any of the individual externalizing disorders were associated with college entrance exam scores or high school graduation, likely due to similar reasons mentioned in the above section on the level effects of internalizing symptoms. In general, the effect sizes for externalizing symptoms were comparable in magnitude to the effects of internalizing symptoms, in contrast to earlier studies showing a stronger effect of externalizing symptoms than internalizing symptoms (e.g., Deighton et al., 2018). Future research should investigate the hypothesized motivational, cognitive, emotional, and behavioral pathways mediating the effect of psychiatric disorders to understand better the differential associations between specific disorders and various indicators of academic achievement.

Psychiatric Disorder Slopes and School Outcomes

Internalizing Symptom Slope Effects

The slope effects were generally counterintuitive and in the opposite direction of the level effects. Specifically, we found that compared with students who experienced slower increases in internalizing symptoms (MDD and GAD in particular) across adolescence, those who experienced faster increases were more likely to succeed in school after controlling for internalizing symptoms levels. These slope effects were larger in magnitude than the level effects. The follow-up sensitivity analyses showed that the effect is robust in the present study. Notably, we found that the association between internalizing symptoms and academic achievement became less negative, and in some cases positive, as the temporal distance of the measurement decreased. The decrease in magnitude and the change in the direction of the association between internalizing symptoms and school outcomes likely led to the positive slope effect. However, this does not explain the effect but points to another counterintuitive finding: Why would having more internalizing symptoms right before and during high school be associated with better school outcomes? Typically, prospective effects get progressively stronger as the time interval between the predictor and the outcome decreases, which in the present context would lead us to expect the negative associations between internalizing symptoms and school outcomes to become even more negative.

One potential explanation is that Mexican-origin youth with a stronger achievement orientation experience more anxiety and depressive symptoms toward the end of high school because of more demanding academic work combined with parental pressure to do well in school while also meeting familial obligations such as caring for younger siblings (East & Hamill, 2013; Wikle et al., 2018) and doing household chores. In other words, a more rapid increase (or slower decrease) in internalizing symptoms may reflect a psychological response, especially for high-achieving students from lower socioeconomic backgrounds to pressures associated with maintaining good grades, fulfilling graduation requirements, and applying to college. Moreover, Mexican Americans, compared with European Americans, tend to have lower mental health literacy and higher mental health stigma (Benuto et al., 2019), which makes Mexican Americans less likely to seek help from mental health professionals and more likely to seek help from family or religious leaders (Gearing et al., 2024). As a result, a Mexican-origin youth's normal psychological responses to stress may persist and accumulate to a maladaptive degree due to a lack of professional mental health support.

Future research should investigate whether the positive association between internalizing symptom slope and school outcomes generalizes to students from other racial–ethnic and socioeconomic backgrounds, particularly those transitioning from high school to college, because students are likely to experience increases in stress as they approach the end of high school.

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Externalizing Symptom Slope Effects

We found that faster increases in externalizing symptoms (ADHD and ODD in particular) predicted higher achievement test scores and a higher likelihood of graduating from high school. To examine the robustness of the slope effect, we conducted the same sensitivity analyses. None of these attempts systematically changed the direction or significance of the slope effect, suggesting that the counterintuitive slope effect is not a methodological artifact in the present sample. However, when conceptually replicating the effect of externalizing symptom slope, we found negative slope effects for the two alternative measures of delinquency and substance use. That is, faster increases in delinquency and substance use predicted worse school outcomes, consistent with prior research and theory. Given that the slope effects of DISC-IV externalizing symptoms were not replicated using the two alternative measures, the positive slope effects of DISC-IV externalizing symptoms do not seem to be robust.

Incremental Validity of Psychiatric Disorders and Temperament

Level Effects

We found strong support for the incremental validity of temperament levels over psychiatric disorder levels in predicting school outcomes, but not vice versa. In other words, temperament levels explained variability in school outcomes after controlling for psychiatric disorder levels. The inclusion of psychiatric disorder levels in the model had minimal impact on the effect sizes of temperament levels. By contrast, the inclusion of temperament levels significantly reduced the effect sizes of psychiatric disorder levels. The incremental predictive validity of temperament levels may be particularly pronounced among students from marginalized backgrounds. Students from immigrant families, racial–ethnic minority communities, and low socioeconomic backgrounds experience chronic stress and unpredictability (Conradt et al., 2020). When striving to develop adaptive functioning strategies despite adversity, students with more adaptive temperament profiles are better positioned to navigate multiple cultural contexts, acquire resources, and obtain support from teachers, all of which are essential for success in a predominantly White educational system. Although students with few psychiatric symptoms may not experience cognitive, emotional, or regulatory deficits, they do not necessarily have an advantage in seeking contextual resources to support their academic pursuits.

Interestingly, a few findings showed a classic suppression effect, wherein the effect sizes of temperament levels increased after controlling for psychiatric disorder levels (e.g., effortful control predicting high school GPA, negative emotionality predicting achievement test scores). Other findings showed a negative suppression effect, such that the effect of psychiatric disorders reversed direction after controlling for temperament domains (e.g., MDD [college attendance] and CD [achievement test scores]). Note that the suppression effect is unlikely due to multicollinearity between temperament and psychiatric disorder levels because the zero-order correlations were smaller than .60 (Supplemental Table S13). Considering that these suppression effects did not emerge consistently across predictors or outcome variables and that these findings were derived from exploratory analysis, these suppression effects should be interpreted with caution. More research is needed to examine whether these suppression effects can be replicated.

In general, our findings suggest that knowing a child's temperament levels will provide more information about their later school achievement than knowing about their psychiatric symptoms. Aligning with National Institute of Mental Health's Research Domain Criteria framework (Cuthbert, 2022; Insel et al., 2010), symptom counts do not fully capture the variance in motivational, cognitive, emotional, behavioral, and interpersonal processes that underpin learning and school achievement. Consequently, relying solely on symptom counts yields weaker and less robust predictions of academic achievement. By contrast, temperament accounts for a more substantial proportion of the variance in the factors that contribute to school outcomes. Therefore, temperament may be a more informative predictor of school outcomes than psychiatric disorders.

Slope Effects

We found that temperament and psychiatric disorder slopes had incremental validity over each other. In other words, the developmental trajectories of temperament and psychiatric disorders, characterizing the pattern of person–environment transactions, had independent effects on school outcomes. In most cases, controlling for psychiatric disorder slopes led to a small reduction in the effect sizes of temperament slopes, whereas controlling for temperament slopes led to a large reduction in the effect sizes of psychiatric disorder slopes. In a few cases wherein high school graduation and college attendance were the focal outcomes, controlling for temperament slopes had a classic suppression effect on psychiatric disorder slopes, increasing the magnitude of the effects. More research is needed to test whether these suppression effects are robust or a statistical artifact when predicting binary outcomes. Together, our findings suggest that the effects of temperament and psychiatric disorders are more distinguishable developmentally than at a static time point. Trajectories of temperament and psychiatric disorders likely capture different aspects of person–environment interactions during adolescence (Hopwood et al., 2022).

Limitations and Future Directions

The present study has several limitations. First, given our passive longitudinal design, we cannot draw conclusions about the causal effects of psychiatric disorders on academic outcomes. Second, our data were not missing completely at random, but rather the missing pattern was associated with demographic variables. Controlling for these variables did not systematically change our findings, but the missing pattern may have been due to variables that were not examined in the present study. Third, because of the zero-inflated distributions of SUDs, we advise against interpreting the effect sizes of SUDs. However, the direction of effects (i.e., negative associations between SUDs and school outcomes) was robust and replicated using an alternative measure of SUDs with less zero inflation. Fourth, the slope effects of externalizing disorders were not robust because we did not replicate the positive slope effects using alternative measures of externalizing disorders.

The findings highlight several avenues for future research. First, it is important to identify the mediating mechanisms that account for some of the level and slope effects, including the cognitive, motivational, and interpersonal processes, as well as the person–environment transactions, discussed in the introduction. Second, the observed associations are likely moderated by a wide range of factors. For example, social and emotional support across the family, peer, and school contexts may provide resilience against academic failure. Contextual support may function either as a moderator that buffers the effects of psychopathology or as a compensatory factor that directly enhances academic achievement and offset the negative effects of psychopathology. Future research should examine these contextual resilience factors to determine how they influence school outcomes in combination with various forms of psychopathology. Finally, future research should replicate the counterintuitive slope effects using other well-validated dimensional measures of psychiatric disorders. Importantly, future research should consider using new measures that provide holistic dimensional assessments of the internalizing disorders, externalizing disorders, and the general psychopathology factor (*p* factor) rather than simply summing symptoms across individual disorders.

Notwithstanding these limitations, the present study highlights the important role that both psychiatric disorder and temperament play in shaping the academic success and failure of Mexican-origin youth, who need to overcome a wide range of systemic and structural barriers to achieve their academic goals.

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